TVAC

Tokenization Value-Add Calculator

INDEPENDENT. STRUCTURED. FOR REAL-WORLD DECISIONS.

Deep Assessment Report

Residential Real Estate SPV Notes — Luxembourg / Germany

A case-specific assessment report for tokenization viability and value creation.

November 2, 2026

A TVAC assessment by Astoval

CLARITY
BEFORE
COMMITMENT.

About this report

TVAC evaluates whether tokenizing a specific asset / structure creates net added value, using the five-factor model:

> **Added Value = (New Opportunities (NO) + Cost Savings (CS) + Risk Reduction (RR)) – (Tokenization Costs (TC) + New Risks (NR))**

Where this report fits

TVAC is designed for the very early decision stage of a tokenization initiative. It tests whether the proposed use of tokenization appears likely to create genuine added value before significant resources are committed to legal structuring, issuance-platform or provider selection, detailed technical architecture, or implementation. A positive verdict is therefore not implementation approval; it indicates that the case may justify deeper legal, commercial, technical and operational work.

Case description summary

The initiating organization is a privately owned real estate holding company owning three stabilized residential assets in Berlin and Hamburg through German GmbHs. The proposed case is intended to raise EUR 15–20 million to finance renovations, partially refinance bank debt, and strengthen liquidity buffers. Instead of raising additional traditional mezzanine capital, the organization is evaluating tokenization through a Luxembourg SPV issuing profit-participating notes linked to net rental cash flow after operating costs and senior debt service. The properties generate steady rental income but require energy upgrades, façade improvements, and selected tenant-area refurbishments to remain competitive and compliant with evolving ESG standards.

The full case description and structured input record used for this assessment are reproduced in Appendix E.

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Methodology	TVAC — Tokenization Value-Add Calculator
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01 Executive summary

TOKENIZATION VALUE VERDICT

Conditional Go

ADDED VALUE

-3.00

plus 8.00 **minus** 11.00

Purpose and tokenization test

Project objective: Raise EUR 15–20 million for energy upgrades, façade improvements, selected tenant-area refurbishments, partial bank-debt refinancing, and liquidity buffers for three stabilized residential assets in Berlin and Hamburg, while testing whether a tokenized Luxembourg note structure is better than additional traditional mezzanine capital for that purpose.

Intended beneficiaries: The immediate beneficiary is the real estate holding company seeking financing flexibility and potentially more modular investor administration. The intended capital providers are professional and other eligible non-retail investors in Germany and the Netherlands who may value smaller ticket access, contractual cash-flow participation, and some post-lock-up transfer optionality. It remains unresolved how much of the benefit would come from the token layer itself rather than from the underlying real-estate strategy and ordinary digital servicing.

Conventional comparator: Additional traditional mezzanine capital or a conventional private-placement note structure used to fund the same German residential refurbishment, partial refinancing, and liquidity-buffer plan without the token layer.

What tokenization must improve: Does a Luxembourg SPV issuing tokenized profit-participating notes improve distribution, transfer control, investor transparency, or long-term servicing enough to justify its extra cost and complexity relative to ordinary mezzanine financing for the same Berlin/Hamburg assets?

ASSESSMENT RATIONALE

The tokenization case scores -3.00 (Conditional Go); this verdict assesses incremental value rather than the underlying asset's investment merits. The token-specific mechanism is a controlled, professionally distributed note with whitelist-based onboarding, regulated custody, contractual profit participation, and planned quarterly liquidity windows after a 12-month lock-up. That could help smaller professional tickets and cleaner transfer governance, but no incremental economic benefit is yet established because venue use, investor demand, tax treatment, and whole-life cost versus conventional mezzanine remain unresolved.

Important interpretation note: This verdict concerns tokenization added value only. A Viable or Conditional Go verdict does not mean that the underlying asset is a good investment, that projected returns are realistic, that credit risk is acceptable, or that the instrument is suitable for any investor.

KEY TAKEAWAYS

01 **Key findings:** Incremental opportunity exists only if smaller professional tickets, transfer controls, and structured post-lock-up trading improve access or investor usability beyond the conventional mezzanine comparator.

02 **Key constraints:** Focus on incremental whole-life resources created by the Luxembourg wrapper, provider stack, venue coordination, and cross-border operating model, and identify who pays them.

03 **Decision posture:** Run a like-for-like design and cost comparison between the proposed tokenized Luxembourg note and a conventional mezzanine/private-placement alternative for the same assets, together with transaction-specific rights, venue, and tax mapping; continue if the token route shows a clear operational or distribution advantage after full provider costs and cross-border frictions are included, redesign if the advantage is narrow but recoverable through scope reduction, and stop if the token layer mainly adds complexity without changing access or servicing outcomes.

04 **Impact:** For the initiating stakeholder, the -3.00 score does not establish positive net added value. A decision to continue requires evidence of sufficient incremental benefits after the token-specific costs and risks.

02 Asset class & market context

This is a rights, ownership-structure, distribution and servicing case. Tokenization should not be treated as creating liquidity by itself; the relevant test is whether the proposed structure improves investor access, administration, governance or transaction processes relative to a conventional SPV, fund or direct-ownership arrangement. The underlying asset's performance and the attractiveness of its cash flows must be separated from the question of how the investment is financed and serviced.

The economically relevant alternative is financing the same issuer and underlying asset through a conventional digitally administered private placement, debt or participation instrument with equivalent rights, investor restrictions, subscriptions and reporting. Digital onboarding, small denominations and electronic registers may also be available without tokenization; they cannot be credited to the token layer without comparison.

Tokenization could change distribution economics only if it demonstrably attracts investors or changes acquisition and servicing costs relative to that digital alternative. Identity-linked transfers and wallet-based holding may alter control and investor support, but technical transferability is not evidence of an active exit market. Issuer and investor benefits must be evaluated after custody, compliance, technology maintenance and long-horizon continuity costs.

Before issuance, test whether investor acquisition, subscription conversion, ongoing servicing and whole-life issuer costs would improve compared with an equivalent conventional digital financing. Without independently verified market evidence, no local funding spread, prevailing investor demand or adoption level can be asserted.

What this context means for the case

Structural factors supporting the case:

- The case presents a real incremental access hypothesis: smaller professional tickets into a Luxembourg-issued cash-flow instrument linked to German residential assets may broaden the reachable investor set beyond a standard mezzanine ticket size.
- The use of a regulated custodian, whitelist-based onboarding, and full KYC/AML provides a controlled operating environment that may reduce some transfer-discipline and recordkeeping risk relative to a loosely managed token model.
- The case describes a legal or economic wrapper that can be assessed for rights mapping, governance and lifecycle administration.
- A controlled investor scope can support a more bounded initial design than unrestricted public distribution.

Structural factors that still constrain the case:

- The structure introduces a meaningful first-issuance burden through the Luxembourg SPV, the tokenized note wrapper, cross-border Germany/Netherlands distribution preparation, regulated custody, and a secondary-transfer operating model.
- The main new risks are wrapper and operating-model risks, especially cross-border tax handling, rights-and-record mapping, venue dependence, and provider-stack coordination.
- Property rights, issuer or SPV rights, authoritative registers and investor entitlements must remain aligned across legal documentation and the token layer.
- Secondary demand, valuation, transfer friction and the full legal and administrative cost stack must be compared with realistic conventional alternatives.
- The multi-jurisdiction sale scope increases the need to separate common design logic from local distribution, investor-category, marketing and transfer requirements.

03 Analysis result

ADDED VALUE

-3.00

PLUS	MINUS
8.00	11.00

Purpose and tokenization test

Potential mechanism and evidence: The token-specific mechanism is a controlled, professionally distributed note with whitelist-based onboarding, regulated custody, contractual profit participation, and planned quarterly liquidity windows after a 12-month lock-up. That could help smaller professional tickets and cleaner transfer governance, but no incremental economic benefit is yet established because venue use, investor demand, tax treatment, and whole-life cost versus conventional mezzanine remain unresolved.

Decision-critical unknowns

- Can the proposed Luxembourg-issued instrument be distributed into Germany and the Netherlands in the intended form without redesigning the investor perimeter, disclosure posture, or venue model, and how would that change sponsor cost and timing?
- Will the intended investor segment actually value smaller professional tickets and quarterly post-lock-up liquidity enough to improve placement economics versus a conventional private placement?
- What are the all-in setup and recurring costs of the tokenized structure compared with additional traditional mezzanine capital for the same financing need, and who bears each incremental cost line?
- How exactly do legal rights, ownership or entitlement records, cash-flow calculations, and payment operations connect from the German GmbHs through the Luxembourg SPV to investors, and where could friction or disputes arise?
- Do cross-border withholding and reporting mechanics materially erode investor returns or sponsor efficiency in a way that would outweigh the stated transparency and access benefits?

Verdict rationale:
This case lands at Conditional Go because the current structure produces an Added Value of -3.00: the strongest supportive driver is New Opportunities (5/10), while the main friction is Tokenization Costs (7/10). Incremental opportunity exists only if smaller professional tickets, transfer controls, and structured post-lock-up trading improve access or investor usability beyond the conventional mezzanine comparator. Focus on incremental whole-life resources created by the Luxembourg wrapper, provider stack, venue coordination, and cross-border operating model, and identify who pays them. This rationale concerns tokenization added value only and does not validate legal, regulatory, tax, financial, investment, credit, yield, issuer, or implementation suitability.

Factor score profile

Factor	Score	Case-specific driver
New Opportunities	**5/10**	Incremental opportunity exists only if smaller professional tickets, transfer controls, and structured post-lock-up trading improve access or investor usability beyond the conventional mezzanine comparator.
Cost Savings	**0/10**	No current credit without evidence that tokenized servicing lowers recurring sponsor burden relative to an ordinary digitally administered private placement.
Risk Reduction	**3/10**	Only additional control benefits count, such as regulated custody and transfer discipline; shared real-estate and financing risks are not token-specific risk reduction.
Tokenization Costs	**7/10**	Focus on incremental whole-life resources created by the Luxembourg wrapper, provider stack, venue coordination, and cross-border operating model, and identify who pays them.
New Risks	**4/10**	Count only additional exposures introduced by the token layer, such as rights-mapping, provider dependence, venue execution, and cross-border payment/tax friction; do not reclassify shared property-market risks as token-specific.

Added Value: -3.00 (plus 8.00 / minus 11.00)

What still needs to be verified:

These are the principal evidence gaps requiring independent verification before relying on this assessment for further decisions.

- No verified case-specific regulatory claim is activated; approved baseline context supports structured scoping, while case-specific legal application remains unresolved.
- Obtain written jurisdiction-specific legal analysis of classification, offering pathway, rights mapping and transfer controls.
- Document and validate the enforceable link between the token, underlying rights, ownership records and cash-flow waterfall.

Regulatory status: Luxembourg · Case-specific applicability unverified

TVAC has identified relevant regulatory context for the stated issuance jurisdiction and case-specific legal questions, but it has not verified how the law applies to the final structure. For an existing or retrospective case, historical authorizations and ongoing obligations require independent validation; for a planned case, applicable requirements should be assessed before implementation.

Sources used and outstanding validation questions are summarized in Appendix A.5.

The score becomes useful only when the structure behind it is coherent. The sections below test the value pathways, participant or investor perimeter, operating burden, regulatory path, custody and transfer fit, and—where material—the settlement and cash-leg design that must support the claimed benefits.

3.1 Value pathways

This section isolates the mechanisms through which tokenization could create incremental value over a realistic conventional alternative. It is not an assessment of whether the underlying asset or financing need is attractive in its own right.

- The proposed tokenized instrument is tied to a concrete financing need: EUR 15–20 million for energy upgrades, façade improvements, selected tenant-area refurbishments, partial bank-debt refinancing, and liquidity buffers across three stabilized residential assets in Berlin and Hamburg.
- The clearest incremental value hypothesis versus additional traditional mezzanine capital is not the underlying real estate itself, but the combination of smaller professional ticket sizes, whitelist-based transfer control, and more standardized investor transparency around rental cash-flow participation.
- Because the notes do not provide operational control over the properties, the investor proposition depends heavily on contractual economics, reporting quality, and confidence in the cash-flow waterfall after operating costs and senior debt service.
- The stated quarterly liquidity windows could add optionality for professional investors, but this benefit remains bounded until the venue route, participation model, and actual order-flow support are independently evidenced.
- Long-term administrative efficiency is a stated objective, but no validated comparator has been supplied to show that tokenized servicing will outperform a conventional digitally administered private placement for this asset owner.

3.2 Investor segmentation

Investor scope affects both upside and friction. A narrower professional or institutional audience can reduce distribution and compliance complexity, but it can also limit the access or liquidity benefit that tokenization is expected to create.

- The intended investor base is professional and other eligible non-retail capital in Germany and the Netherlands; retail participation is explicitly out of scope.
- That segmentation aligns with the absence of operational control rights and with the underlying exposure to German residential rental cash flows plus refurbishment execution risk.
- The interface label 'accredited' should be treated as a generic user label rather than a confirmed EU legal category; the final investor-category mapping in Germany and the Netherlands remains to be confirmed.
- Smaller professional tickets are part of the stated value case, but the economics depend on whether onboarding, reporting, and transfer controls can serve more investors without pushing sponsor-side operating costs above the benefits.
- Investors will likely focus on cash-flow reporting, senior debt interaction, refurbishment governance, and exit mechanics more than on the token format itself.

3.3 Operational costs

This section distinguishes plausible process savings from the additional cost of introducing the token layer. A saving becomes a strong value driver only when it can be compared with the current process and is not offset by data, integration, provider or compliance work.

- The cost burden is likely to be front-loaded because the structure adds a Luxembourg SPV layer, cross-border distribution work, custody, venue coordination, onboarding controls, and rights documentation on top of the underlying real-estate financing process.
- Ongoing cost categories are also likely to remain relevant: investor registry or recordkeeping functions, periodic reporting, distribution processing, venue-related coordination, compliance operations, and provider oversight.
- No evidence has been supplied that these added costs are offset by lower fundraising friction or materially lower servicing cost than a conventional private placement or mezzanine note for the same Berlin/Hamburg renovation strategy.
- The administrative-efficiency thesis may become stronger if the same issuance stack is reused for repeat financings, but that scale effect has not been demonstrated for this specific transaction.
- Provider-pricing asymmetry is a key concern for the Asset Owner: if custodians, venues, and issuance-service providers are paid regardless of whether investor access or servicing efficiencies improve, value capture can shift away from the sponsor.

3.4 Regulatory path

This is regulatory scoping, not a legal opinion. The purpose is to identify the verified regulatory context and the design questions that materially affect the case, while making clear what still requires qualified jurisdiction-specific validation.

- TVAC has approved baseline regulatory context for Luxembourg, but it has not activated a verified case-specific local claim. The frameworks below are used for structured scoping only; applicability and current detailed conditions remain to be independently confirmed.
- EU scoping should first distinguish a token that is a financial instrument or other regulated financial product from a crypto-asset governed by MiCA. MiCA does not replace the EU financial-services framework for instruments already covered by that framework, and tokenization does not itself determine classification.
- Where the token represents a transferable security or another financial instrument, MiFID II, the Prospectus Regulation and related national rules may shape distribution, investor access, intermediary roles and disclosure. The final facts determine which requirements or exemptions are relevant.
- The DLT Pilot Regime provides an EU framework for specified DLT market infrastructures, while MiFID II, CSDR and national implementation remain relevant to trading,

settlement, custody, recordkeeping and market-infrastructure functions according to the final design.

- EU-level analysis must be combined with member-state law and supervisory practice for issuer or SPV law, property and security rights, authoritative registers, insolvency, tax, marketing and operational roles. An EU framework reference does not resolve those national questions.
- For real-estate tokenization, the financial-instrument and distribution analysis must be separated from the national legal treatment of the property, the issuer or SPV, investor rights and land or ownership records. The token layer should not be assumed to replace legally authoritative national records.
- Approved baseline context identifies the following potentially relevant frameworks or regimes for scoping in Luxembourg: EU securities and markets framework (MiFID II / Prospectus / etc.) (EU baseline); EU crypto-asset framework (MiCA); EU fund/management frameworks (as relevant); AML regime (Luxembourg / EU); Data protection (GDPR). Their case-specific applicability, current conditions and interaction must be independently confirmed.
- Issuer structure and investor base often determine the dominant regulatory path.
- Distribution/marketing constraints can be a primary design driver for cross-border setups.
- Custody/safeguarding and governance controls can be decisive for institutional participation.
- Germany: approved baseline scoping context is available for distribution, investor eligibility, marketing, transfer and intermediary-role analysis. It does not determine local applicability or compliance.
- Germany — baseline scoping: Approved baseline context identifies the following potentially relevant frameworks or regimes for scoping in Germany: EU securities and markets framework (MiFID II / Prospectus / etc.) (EU baseline); German electronic securities framework (general); AML regime (Germany / EU); Data protection (GDPR). Their case-specific applicability, current conditions and interaction must be independently confirmed.
- Germany — baseline scoping: EU baseline applies; local supervisory practice and implementation details can matter.
- Germany — baseline scoping: Tokenized instruments that resemble securities typically inherit securities offering/distribution constraints.
- Netherlands: approved baseline scoping context is available for distribution, investor eligibility, marketing, transfer and intermediary-role analysis. It does not determine local applicability or compliance.
- Netherlands — baseline scoping: Approved baseline context identifies the following potentially relevant frameworks or regimes for scoping in Netherlands: EU securities and markets framework (MiFID II / Prospectus / etc.) (EU baseline); EU crypto-asset framework (MiCA); AML regime (Netherlands / EU); Data protection (GDPR). Their case-specific applicability, current conditions and interaction must be independently confirmed.
- Netherlands — baseline scoping: Classification and distribution perimeter remain the main gating factors.
- Netherlands — baseline scoping: Marketing constraints and investor communication discipline matter for compliance outcomes.

3.5 Custody & venue fit

Custody and venue design should be proportionate to the value thesis. If the case does not depend on continuous secondary trading, extra custody, venue or transfer complexity should be justified by a concrete operational, control or distribution benefit.

- A regulated custodian is a strong fit for the stated professional-investor strategy because it may support institutional comfort, controlled onboarding, and cleaner segregation of wallet or account responsibilities than a self-custody design.
- The venue concept is directionally compatible with the case because the issuer does not promise quarterly liquidity and instead proposes quarterly liquidity windows after a lock-up, which is more realistic for a tokenized real-estate cash-flow instrument.
- However, the user-selected venue label 'MTF or other regulated venue (user-selected interface label; local classification unconfirmed)' is not yet an evidenced operating route in EU case-specific terms; the final venue model, access rules, and responsibility split between the venue, custodian, and issuer remain unresolved.
- The fit between custody and venue will matter operationally because transfers, investor eligibility, lock-up controls, and distribution recordkeeping must work across the same operating stack without creating manual reconciliation that offsets the stated efficiency thesis.
- If the venue remains only a design target and not a contracted pathway, the liquidity proposition stays a potential benefit rather than an evidenced one.

3.6 Settlement & Cash-Leg Readiness

Settlement is assessed against the case's actual value logic, not against an assumption that tokenization requires digital money or T+0. The key question is whether the cash leg can operate reliably for the intended investors and operating model without unnecessary complexity.

- Why it matters: A material part of the claimed value depends on the settlement design working reliably for the intended participants.
- Current position: the proposed payment and settlement approach is T+1/T+2 securities settlement. A settlement approach has been proposed, but its operational readiness has not been independently established. This does not establish legal or operational settlement finality.
- What must be proven: asset- and cash-leg finality, synchronization, failed-settlement handling, exception management and recovery. Until those points are evidenced, settlement-dependent benefits should remain conditional rather than treated as realized outcomes.

04 Stakeholder value & alignment

This chapter asks whether the parties needed to make the structure work are economically and operationally aligned. It focuses on the value each party receives, the burden and responsibility it carries, and the dependencies that could weaken the project's overall Added Value.

4.1 Value & alignment overview

The stakeholder map identifies the parties involved. Their distinct benefits, costs, responsibilities and alignment conditions are analysed once in the sections that follow.

Privately owned real estate holding company / issuance sponsor

Owner of three stabilized residential assets in Berlin and Hamburg seeking refurbishment and balance-sheet financing

Professional and other eligible non-retail investors in Germany and the Netherlands

Intended holders of profit-participating notes linked to net rental cash flow after operating costs and senior debt service

Custody, onboarding, venue, and SPV-related service providers

Delivery counterparties needed to operationalize the Luxembourg tokenized note structure

4.2 Privately owned real estate holding company / issuance sponsor

Value received

- Potentially gains access to a more modular professional-investor product with smaller ticket sizing, controlled transferability, and a more standardized reporting proposition.

Costs, risks & responsibilities

- Carries most of the first-issuance complexity, including Luxembourg SPV setup, provider coordination, tax and rights mapping, venue design, and the risk that administrative efficiencies do not outweigh added costs.

Alignment & actions

- Alignment improves only if the token stack measurably helps fundraising or servicing; otherwise the sponsor bears disproportionate cost while infrastructure providers still earn fees.

4.3 Professional and other eligible non-retail investors in Germany and the Netherlands

Value received

- May receive cleaner contractual economics, regulated custody, controlled onboarding, and some optionality from quarterly liquidity windows after the lock-up.

Costs, risks & responsibilities

- Remain exposed to refurbishment execution, cash-flow variability, senior debt interaction, tax treatment, and the possibility that secondary transfer options are more limited in practice than in concept.

Alignment & actions

- Alignment is strongest if investor reporting, cash-flow calculations, transfer mechanics, and rights documentation are precise and operationally reliable.

4.4 Custody, onboarding, venue, and SPV-related service providers

Value received

- Benefit from setup fees, recurring servicing revenue, and a defined operating role if the structure launches and remains active.

Costs, risks & responsibilities

- Must deliver coordinated onboarding, recordkeeping, settlement support, transfer controls, and operational continuity across jurisdictions.

Alignment & actions

- Alignment is incomplete until pricing, responsibilities, portability, and performance standards are contractually tied to the actual value delivered to the sponsor and investors.

4.5 Who bears the costs and risks?

In a capital-formation case, the issuer may carry substantial structuring, legal, compliance and distribution costs before funding is secured. That is a pre-funding economic position, not necessarily the final outcome. The relevant question is whether tokenization demonstrably produces incremental capital access, financing flexibility or terms that outweigh the added stack relative to a realistic conventional alternative.

- A materially negative pre-funding issuer position should trigger a demand, terms and cost-allocation review, not an automatic conclusion that tokenization can never work. It becomes a warning sign if investor demand remains uncommitted, the same capital could reasonably be raised conventionally on similar terms, or provider fees absorb the incremental financing benefit.
- If investor net position is low or negative, expect weak demand unless enforceability, disclosures, custody, and realistic exit paths are improved.

05 Factor breakdown – detailed explanation

This chapter explains the causal basis of the five scores: which facts, assumptions and evidence gaps create upside, and which ones introduce additional friction. The three positive factors ask what tokenization adds relative to a realistic conventional alternative; the two negative factors ask what extra cost or risk the token layer introduces. A low positive score therefore means limited or weakly evidenced tokenization benefit, while a low negative score means limited additional friction. The scores are calibrated methodology judgments, not measured outcomes or legal conclusions.

> **Added Value = (New Opportunities + Cost Savings + Risk Reduction) – (Tokenization Costs + New Risks)**

Conditional Go identifies a design-dependent assessment, not a positive implementation decision. Where the aggregate score is negative, the current design has not established net tokenization value. Further work must demonstrate incremental benefits or lower costs and risks against the stated conventional comparator before a rollout is justified. With an Added Value score of -3, the case remains condition-sensitive and should be judged by whether the key gating conditions can be locked in. Net Added Value is primarily driven by new opportunities at 5/10 on the upside, and constrained mainly by tokenization costs at 7/10 on the friction side.

New opportunities (NO)

- The case presents a real incremental access hypothesis: smaller professional tickets into a Luxembourg-issued cash-flow instrument linked to German residential assets may broaden the reachable investor set beyond a standard mezzanine ticket size.
- Whitelist-based transfer controls, a regulated custodian, and planned quarterly liquidity windows after a 12-month lock-up provide a more structured transfer path than a purely static private note, which supports the opportunity thesis.
- The upside remains capped because Germany and the Netherlands are the only stated sale jurisdictions, retail investors are excluded, and no venue commitment or investor-demand evidence has been supplied.
- The opportunity case therefore rests on plausible distribution and transfer improvements, not on proven liquidity or a broad market-expansion outcome.

Cost savings (CS)

- Administrative efficiency is stated as a long-term objective, but no validated baseline has been supplied for the current cost of raising and servicing equivalent traditional mezzanine capital.
- The proposed structure adds sponsor-side coordination across a Luxembourg SPV, German property-holding entities, regulated custody, onboarding controls, and venue-related functions, which prevents a current claim of net recurring savings.
- Any future efficiency would likely depend on repeat use of the same issuance stack or genuinely lower servicing effort per investor, and that has not been demonstrated for this transaction.

- For this reason, the current assessment does not credit cost savings simply because parts of the investor administration process may be digitized.

Risk reduction (RR)

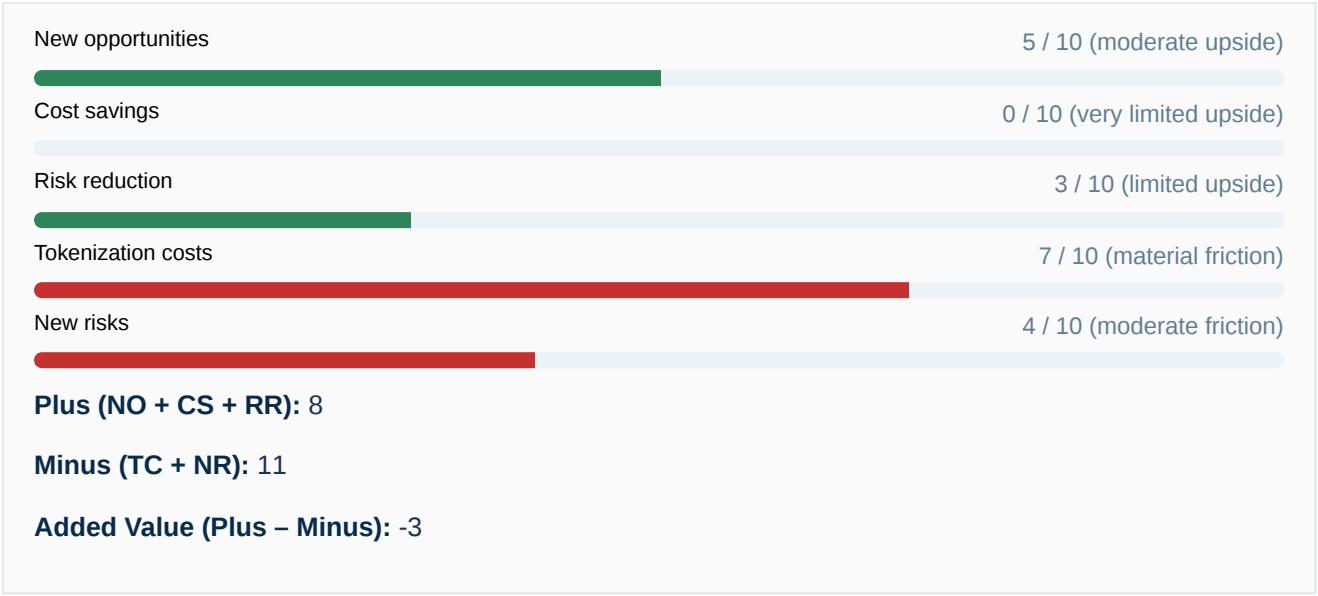
- The use of a regulated custodian, whitelist-based onboarding, and full KYC/AML provides a controlled operating environment that may reduce some transfer-discipline and recordkeeping risk relative to a loosely managed token model.
- The absence of property-level operational control rights for investors can simplify governance boundaries because the instrument is framed as economic participation rather than shared asset management.
- Improved investor transparency is a plausible benefit if reporting is well designed, especially where profit participation depends on rental cash flow after operating costs and senior debt service.
- The reduction is limited because the legal authority of the records, the full rights chain, and the cross-border operating treatment are not yet independently confirmed.

Tokenization costs (TC)

- The structure introduces a meaningful first-issuance burden through the Luxembourg SPV, the tokenized note wrapper, cross-border Germany/Netherlands distribution preparation, regulated custody, and a secondary-transfer operating model.
- Those costs are not merely technical; they also include documentation, provider management, investor onboarding discipline, and ongoing coordination between the note structure and the underlying German GmbH asset owners.
- The issuer has not yet shown that this whole-life cost stack is lower than a conventional non-tokenized mezzanine or private-placement alternative for the same financing need.
- Because the transaction is framed as an evaluated first structure rather than a proven repeat program, the fixed setup burden weighs heavily on the current case.

New risks (NR)

- The main new risks are wrapper and operating-model risks, especially cross-border tax handling, rights-and-record mapping, venue dependence, and provider-stack coordination.
- The intended liquidity feature introduces an execution risk of its own: if quarterly windows exist in design but not in meaningful participation, investor expectations and sponsor effort can diverge.
- The contractual note structure linked to German rental cash flows through a Luxembourg SPV requires clean legal-technical mapping; if that mapping is weak, disputes or operational friction can arise even when the underlying properties perform.
- By contrast, renovation execution risk, valuation stability, and interest-rate sensitivity are mainly underlying asset or financing risks that also exist in conventional structures, so they do not by themselves justify a higher token-specific risk assessment.



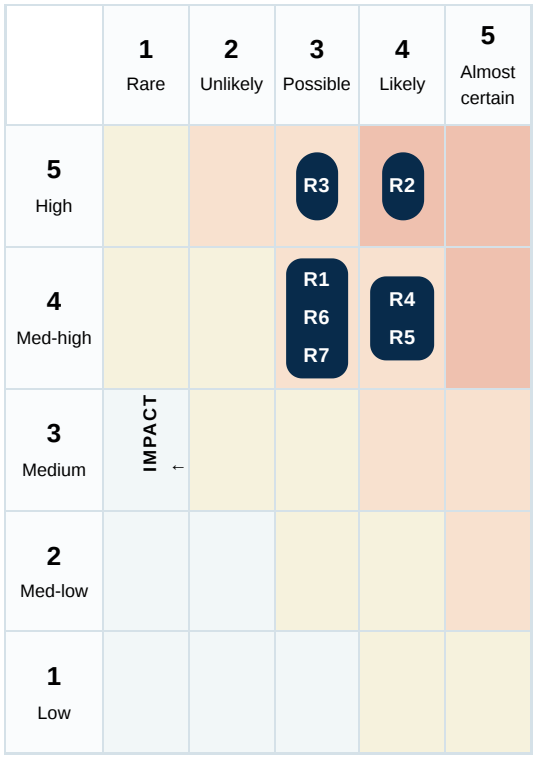
Plus (NO + CS + RR): 8

Minus (TC + NR): 11

Added Value (Plus – Minus): -3

06 Key risks & mitigation

This matrix prioritizes the report's identified risks by impact and likelihood. It is not a measured probability model; each position should be read together with the case-specific description and mitigation.



Risk zones are indicative prioritization bands, not measured probabilities.

PRIORITY RISKS

R1 **Misaligned incentives / vendor-capture risk**

Impact 4 · Likelihood 3

Mitigation: Use stage-gating and milestones (legal opinions + demand validation) before committing; negotiate fee structures with meaningful success-based components; ensure portability/exit clauses if the platform/venue changes strategy.

R2 **Cross-border tax and payment-friction risk**

Impact 5 · Likelihood 4

Mitigation: Obtain independent tax-operating mapping for cash flows from the German GmbHs through the Luxembourg SPV to investors in Germany and the Netherlands, including withholding, reporting, and exception handling.

R3 **Rights-and-record mismatch risk**

Impact 5 · Likelihood 3

Mitigation: Document the full rights chain from token holder to noteholder to SPV claim to underlying rental cash-flow entitlement, and identify which records control distributions and transfers in the final design.

R4 **Venue and liquidity-underperformance risk**

Impact 4 · Likelihood 4

Mitigation: Validate the post-lock-up venue pathway, participant access model, and expected operating process for quarterly liquidity windows before relying on liquidity as a selling point.

R5 **Provider-stack cost overhang**

Impact 4 · Likelihood 4

Mitigation: Secure binding pricing, service-level, portability, and termination terms across custody, onboarding, and venue-related providers before finalizing the issuance route.

R6 **Refurbishment execution slippage**

Impact 4 · Likelihood 3

Mitigation: Tie draw planning, reporting, and contingency governance to the actual renovation program so that investor communications reflect cost, timing, and occupancy effects.

R7 **Interest-rate and refinancing pressure**

Impact 4 · Likelihood 3

Mitigation: Model the interaction between senior debt service, liquidity buffers, and profit participation under adverse rate and valuation scenarios before fixing note economics.

07 Recommendations & next steps

7.1 Design priorities and decision gates

The decisive trade-off in this case is clear: the Luxembourg issuance layer, cross-border Germany/Netherlands distribution plan, regulated custody, and planned venue access may create a cleaner professional-investor product for the Berlin/Hamburg residential refurbishment strategy, but they also load significant cost and execution burden onto the Asset Owner before incremental fundraising or servicing benefits are proven. Overall verdict: Conditional Go. The current unresolved decision is therefore not whether the underlying real-estate financing need is credible; it is whether the token layer improves that financing route enough to justify its added complexity for the sponsor. This is a structured concept rather than a fully locked design: the investor perimeter is fairly clear, but the final rights mapping, venue pathway, provider economics, and tax treatment still determine whether the structure is rational for the Asset Owner.

1. Shift provider cost and execution asymmetry away from the Asset Owner

Constraint in this case

The sponsor appears to bear most of the fixed burden of the token stack: Luxembourg SPV setup, cross-border structuring, onboarding controls, regulated custody, venue coordination, and the operating effort needed to support quarterly liquidity windows. If providers monetize that stack regardless of whether investor reach or servicing efficiency improves, the Asset Owner absorbs downside while others collect fees.

Required actions

- Obtain binding commercial scopes from the custodian, venue-related counterparties, onboarding providers, and any registry or transfer-administration providers.
- Separate one-off setup charges from recurring servicing charges and map each cost line against the conventional mezzanine comparator.
- Require clear portability, termination, recovery, and handover terms so the sponsor is not locked into a costly operating stack if the first issuance underperforms.
- Tie any optional liquidity-support or secondary-market operating features to demonstrated investor use rather than paying for the full feature set upfront.

Expected effect

This would directly test whether the token layer improves economics for the Asset Owner rather than only creating fee income for infrastructure providers. If commercial terms become more variable, portable, and evidence-linked, the burden on Tokenization Costs could fall and some New Risks tied to vendor

dependence could also ease. Confirmation would come from a like-for-like comparison showing that token-specific fixed and recurring costs are proportionate to the expected investor-access or servicing gains.

2. Turn quarterly liquidity from a selling point into an evidenced operating path

Constraint in this case

Quarterly liquidity windows after a 12-month lock-up are more realistic than a constant-liquidity promise, but they remain an assumption until the sponsor can show how investors actually enter, hold, and transfer the notes. Without a concrete post-lock-up operating route, liquidity becomes marketing language rather than a durable advantage.

Required actions

- Define the exact post-lock-up transfer process across the venue, custodian, whitelist controls, and investor-eligibility checks.
- Document who is responsible for onboarding new eligible buyers, handling failed or disputed transfers, and maintaining synchronized investor records.
- Test whether the intended smaller-ticket professional investor segment values periodic tradability enough to influence demand or pricing.
- Remove or down-rank any liquidity claims in investor materials that cannot be supported by documented venue and operating arrangements.

Expected effect

If the quarterly-window model is operationally real and valued by the intended investor base, it strengthens the opportunity case without relying on an implausible always-on market. If it is weak or unused, the sponsor can simplify the structure and avoid paying for complexity that does not improve fundraising. Evidence that would matter here is not a theoretical feature list, but documented operating responsibilities and investor feedback showing that the transfer option changes behavior or improves placement.

3. Sequence jurisdictions and investor scope more narrowly at launch

Constraint in this case

The current concept spans Luxembourg issuance with sales into Germany and the Netherlands for professional and other eligible non-retail investors. That may be strategically sensible over time, but it raises early legal-scoping, marketing-discipline, tax-operating, and provider-coordination burden before the first issuance has proven its economics.

Required actions

- Stage the rollout by sale jurisdiction rather than assuming the full Germany/Netherlands perimeter must be activated immediately.
- Keep transfer restrictions, whitelist logic, and investor communications tightly aligned with the actual first-phase eligibility assumptions.
- Confirm whether the same investor-access objective can be met with a narrower initial perimeter and a later expansion only if the first issuance performs as intended.
- Preserve the no-retail boundary unless and until the sponsor has evidence that broader access is necessary and supportable.

Expected effect

A narrower first phase could reduce execution burden and avoid paying for unused distribution reach. That would primarily improve the cost and risk side of the Added Value equation, while still preserving the possibility of cross-border expansion if it proves necessary. The confirming evidence would be a jurisdiction-by-jurisdiction design showing that the first launch can meet its fundraising objective without carrying avoidable complexity from day one.

4. Tighten the rights, tax, and record chain from the Luxembourg notes to the German asset cash flows

Constraint in this case

The economic story depends on profit-participating notes linked to net rental cash flow after operating costs and senior debt service, but the strength of that story depends on exact legal and operational linkage. If the rights chain, authoritative records, payment waterfall, or cross-border withholding treatment are unclear, both investor confidence and sponsor efficiency suffer.

Required actions

- Produce a full transaction map from the German GmbHs and property cash flows to the Luxembourg SPV obligations and then to investor-level payments and principal repayment.
- Identify which records drive ownership, distributions, transfer eligibility, and corporate-action processing in the final operating model.
- Confirm how tax withholding, reporting, and reconciliation would work for investors in Germany and the Netherlands under the proposed payment flows.
- Make sure investor disclosure explains the economic rights precisely, including what investors do and do not control at property level.

Expected effect

A cleaner rights-and-record chain would support the transparency thesis, reduce avoidable operating friction, and strengthen confidence that the token is more than a cosmetic wrapper. That may improve the opportunity and risk profile at the same time, but only if the sponsor can evidence that reporting, payments, and transfer records will function consistently across the full structure. The relevant proof is coherent transaction documentation and operating design, not general statements that rights are 'clear'.

Expected Outcome After Restructuring

If the sponsor can narrow the initial scope, contract the provider stack on terms that protect Asset Owner economics, evidence a workable quarterly-transfer process, and document the rights-and-tax chain cleanly, the case would become materially stronger. The underlying Berlin and Hamburg residential refurbishment story is already understandable; the challenge is proving that the token layer improves financing and servicing outcomes rather than simply adding another structured wrapper.

Under revised assumptions, these changes could shift the Added Value profile and the verdict outcome — but this depends on execution, jurisdiction-specific constraints, evidence of the stated participant or user outcomes, and provider commitments. Until those points are validated, the prudent approach is to treat tokenization here as a potentially useful financing architecture that still has to earn its place against conventional mezzanine capital.

7.2 Design what-if scenarios

Does a Luxembourg SPV issuing tokenized profit-participating notes improve distribution, transfer control, investor transparency, or long-term servicing enough to justify its extra cost and complexity relative to ordinary mezzanine financing for the same Berlin/Hamburg assets?

- If the first issuance were staged through only one of the stated sale jurisdictions rather than both Germany and the Netherlands, the distribution hypothesis would narrow but the initial execution burden could also fall materially; that would test whether the wider cross-border perimeter is truly value-creating or mainly complexity-creating.
- If the quarterly liquidity feature remained in investor materials but the venue pathway and operating responsibilities stayed unresolved, the token layer would keep much of its cost and risk burden while part of the opportunity thesis would weaken.
- If the same investor-transparency and reporting outcomes could be delivered through a conventional digitally administered mezzanine structure for the same Berlin/Hamburg assets, the token-specific case would become much harder to justify on incremental value grounds.

7.3 Suggested next steps

- Obtain independent Luxembourg, German, and Dutch scoping on the proposed Luxembourg SPV profit-participating note, the intended professional-investor distribution model, and the user-selected disclosure posture, because the current case depends on whether the structure can be executed without major redesign of opportunity, cost, and risk assumptions, and continue only if at least one workable issuance-and-distribution path is confirmed or the redesign required is commercially acceptable.

- Build a like-for-like comparator between the proposed tokenized note and additional traditional mezzanine capital for the same Berlin/Hamburg refurbishment financing, because the current assessment cannot credit administrative efficiency or fundraising improvement without a conventional baseline for NO, CS, and TC, and continue only if the token route shows a decision-relevant advantage in investor reach, servicing burden, or flexibility after including all token-specific costs.

- Document the full rights, records, and cash-flow chain from the German GmbHs through the Luxembourg SPV to investor payments, because the transparency and risk-mitigation claims depend on enforceable and operable linkage rather than headline structure, and continue only if the final map shows a coherent payment waterfall, recordkeeping logic, and exception-handling process that do not rely on unresolved manual workarounds.

- Validate the quarterly-liquidity design with the intended custodian and venue pathway, because New Opportunities depend on an operating transfer route rather than an aspirational secondary-market statement, and continue only if the post-lock-up process, investor-eligibility controls, and record synchronization can be documented end to end.

- Obtain transaction-specific tax-operating analysis for withholding, reporting, and reconciliation across Luxembourg, Germany, and the Netherlands, because cross-border payment friction can materially alter sponsor economics and investor experience through TC and NR, and continue only if the tax and payment flows remain workable without undermining the intended net-cash-flow participation model.

TVAC

Appendices

- Appendix A — Method & interpretation
- Appendix B — Operational roadmap
- Appendix C — TVAC Provider Directory (illustrative)
- Appendix D — Glossary & abbreviations
- Appendix E — Case inputs used for assessment
- Appendix F — Disclaimer

Appendix A — Method & interpretation

A.1 How to use this appendix

This appendix explains how to interpret the assessment, what evidence is still needed, and which conditions could strengthen or weaken the case. It is not a technical manual for TVAC.

A.2 What the assessment establishes — and what it does not

This report provides informational decision support only. It does not constitute legal, regulatory, tax, financial, or investment advice, and it does not provide a compliance determination, valuation, or provider endorsement.

Case snapshot (as stated)

- Issuance jurisdiction: Luxembourg
- Sale jurisdictions / target markets: Germany, Netherlands
- Asset / instrument type (indicative): wrapper
- Liquidity intent (as stated): quarterly
- Payment / cash-leg indication: T+1/T+2 securities settlement
- Cash-leg dependency / readiness: High dependency / Selected / Unverified readiness

What the Tokenization Value Verdict means

- Viable: Tokenization added value is plausible under the stated assumptions, subject to normal verification and execution.
- Conditional Go: Tokenization added value is design-dependent. Proceeding without meeting the listed conditions materially increases execution risk.
- Non-viable: Under current assumptions, tokenization is unlikely to outperform a conventional structure on a risk-adjusted basis.

What TVAC did not assess

- No definitive legal, regulatory, tax, or investment conclusions
- No asset valuation, yield projections, or market-price forecasts
- No assurance of secondary-market liquidity or investor demand
- No operational readiness audit beyond the indicative checklist

A.3 Minimum evidence before the next major commitment

Before proceeding toward implementation, decision-makers should be able to substantiate a minimum level of evidence across the areas below.

This checklist is not exhaustive and does not imply regulatory sufficiency. It reflects common governance expectations in institutional-grade projects.

Highest-priority gaps in this case

(derived from inputs/completeness)

1. No verified case-specific regulatory claim is activated; approved baseline context supports structured scoping, while case-specific legal application remains unresolved.
2. Obtain written jurisdiction-specific legal analysis of classification, offering pathway, rights mapping and transfer controls.
3. Document and validate the enforceable link between the token, underlying rights, ownership records and cash-flow waterfall.

Structural & economic foundations

- Clear description of the underlying asset and ownership structure
- Defined economic rights and cash-flow logic
- Draft term sheet or structural outline
- Clear rationale for tokenization vs alternatives

Regulatory & distribution posture

- Issuance jurisdiction clearly defined
- Explicit list of sale jurisdictions
- Documented investor type assumptions
- Initial regulatory scoping (high-level)

Token & control design

- High-level token design and rights mapping
- Transfer restriction and whitelist logic (conceptual)
- Corporate actions and investor reporting approach
- Governance for smart-contract or protocol changes (if relevant)

Infrastructure & counterparties

- Custody model (conceptual)
- Indicative partner shortlist (non-binding)
- Issuance, registry, and lifecycle approach
- Payment / cash-leg and settlement approach, proportionate to the case's actual value proposition
- Early assessment of secondary trading (if any)

Cost & operational realism

- High-level CAPEX and OPEX view
- Identification of major cost drivers
- Clear internal ownership of ongoing operations
- Understanding of scalable vs non-scalable costs

Lack of clarity across several of these areas materially increases execution risk, even where the conceptual case appears attractive.

A.4 What could strengthen or weaken the case

Top drivers in this case (derived from scoring)

- Upside driver: New opportunities (value depends on distribution and product/instrument design)
- Risk driver: New risks (most sensitive to regulatory interpretation, custody/ops complexity, and cross-border distribution)
- Cost driver: Tokenization costs (most sensitive to legal/compliance scope, integrations, and ongoing ops overhead)

High-impact decision levers (practical)

- Distribution scope: reduce jurisdictions and/or tighten investor segments to lower compliance and operational overhead.
- Liquidity posture: treat early secondary trading as optional; defer until structure and reporting are stable.
- Custody & controls: choose a control model that meets institutional requirements without over-engineering.
- Payment / cash-leg: clarify the settlement model only to the level required by the claimed value proposition; do not over-engineer it for low-dependency cases.
- Rights & governance: tighten rights mapping (cash flows, reporting, governance, transfer rules) to reduce friction and disputes.

How to use this

If you need the verdict to improve, change the few levers above rather than adding features. In many cases, viability increases by reducing ambition and complexity.

A.5 Evidence & source transparency

Basis of assessment

The assessment is based on the case description and structured inputs used for this assessment, the fixed TVAC methodology, and any activated verified external sources. Operational and commercial claims in the case inputs have not been independently verified unless explicitly stated.

External sources used

- Directive 2014/65/EU on markets in financial instruments · European Union · checked 2026-07-15
- Regulation (EU) 2017/1129 — Prospectus Regulation · European Union · checked 2026-07-15
- Regulation (EU) 2023/1114 — Markets in Crypto-assets · European Union · checked 2026-07-15
- Regulation (EU) 2022/858 — DLT Pilot Regime · European Union · checked 2026-07-15
- Regulation (EU) No 909/2014 — Central Securities Depositories Regulation · European Union · checked 2026-07-15

What remains unverified

- Which offering/distribution posture is intended in each sale jurisdiction, and is the operating model aligned with that posture?
- Which regulated roles (onboarding, custody, venue, agent/registrar) are expected or preferred for the chosen design?
- If secondary trading is planned, what is the feasible venue pathway for the target markets?

Appendix B — Operational roadmap

This roadmap translates the assessment into a staged implementation path for the specific case type. It is not a project plan or implementation instruction: each phase should be adapted to the final legal structure, counterparties, providers and evidence available. Progress only when the decision gate for the preceding phase has been met.

B.1 Operational roadmap (structured)

The items below are indicative planning scaffolding. They are not a substitute for jurisdiction-specific legal, tax, financial, and operational due diligence, and they must be tailored to the selected asset type, wrapper structure, investor segment, custody model, venue, and settlement design.

B.2 Phase 1 — Preparation & feasibility

Case-specific focus for this phase: Primary issuance jurisdiction: Luxembourg. Initial distribution scope: Germany, Netherlands. Initial investor focus: professional investors, other eligible non-retail investors (local classification unconfirmed).

Objectives

- Clarify the underlying asset/instrument, economic rights, and proposed token wrapper.
- Define target investor segments (professional / eligible / retail, if applicable) and distribution scope.
- Establish internal alignment on objectives, success criteria, and risk appetite.

Key decisions

- Token rights model (equity-like, debt-like, fund-like, revenue share, participation note, etc.).
- Transfer restriction regime (whitelisting, holding periods, jurisdictional gating).
- Custody and settlement direction (account-based vs. wallet-based; T+0 vs. T+1+).

Deliverables (checklist)

- One-page case definition: asset, wrapper, jurisdiction(s), investor segment, and target size.
- Preliminary term sheet outline (rights, restrictions, fees, lifecycle events, disclosures).
- High-level implementation estimate (time bands, cost bands, key dependencies).
- Initial risk register (top 10–15 risks with owners and first mitigations).

Primary stakeholders

- Issuer/asset owner (business sponsor, finance, legal/compliance, operations/IT).
- External legal counsel (securities, funds, structuring), tax adviser, and (if relevant) auditor.
- Shortlisted tokenization platform/technology provider, regulated custodian, and potential venue(s).

Dependencies / gates

- Clear legal characterization of the instrument and its regulatory perimeter in the issuance jurisdiction.
- Agreement on must-have constraints (investor eligibility, reporting, custody, settlement, data handling).
- Internal approval to proceed to structured design (budget and governance gate).

Risk hotspots

- Unclear instrument classification or incompatible distribution intent (e.g., retail scope without a verified pathway).
- Mismatch between user-stated transfer expectations and the legally and operationally supportable process.
- Hidden operational burdens (cap-table/admin, reporting, corporate actions, tax documentation).

B.3 Phase 2 — Legal, tax & regulatory structuring

Case-specific focus for this phase: Core legal and tax analysis should start from Luxembourg. Distribution permissions must be validated for Germany, Netherlands. Proposed disclosure approach: Exempt. Applicable local requirements must be confirmed. Transfer-control design to validate: Whitelist + EU gating + 12-month lock-up.

Objectives

- Obtain independent legal validation of a legally supportable issuance and distribution structure for the intended jurisdiction(s).
- Define disclosure, governance, and ongoing compliance obligations for the lifecycle of the instrument.
- Obtain qualified tax analysis of the proposed wrapper and cash-flow mechanics for issuer and investors.

Key decisions

- Issuance vehicle choice (direct issuer vs. SPV; fund wrapper vs. corporate wrapper).
- Offering and documentation pathway verified for the stated jurisdiction(s); do not infer a legal category from interface labels.
- Investor onboarding model (KYC/AML responsibilities, reliance model, delegated vs. in-house).
- Choice of legal documentation stack verified by qualified local advisers.

Deliverables (checklist)

- Legal characterization memo and jurisdiction matrix (requirements by country/investor segment).
- Draft offering documentation and investor disclosures.
- Transfer restriction policy and rulebook (whitelist logic, jurisdictional gating, and any explicitly selected holding-period rules).
- Tax memo(s) covering withholding, reporting, and cross-border distribution considerations.
- Compliance operating model (roles, escalation paths, record-keeping, audit trail).

Primary stakeholders

- Issuer legal/compliance, external counsel, tax advisers.
- Custodian/compliance function (if regulated custody is used), transfer agent (if applicable).
- Any transfer provider or environment explicitly retained in scope and independently validated.

Dependencies / gates

- Independent legal confirmation of the offering pathway and distribution permissions across the stated target jurisdictions.
- Operational feasibility of investor onboarding and transfer controls (including data privacy constraints).
- Formal sign-off on documentation pack and compliance model.

Risk hotspots

- Jurisdiction conflicts (marketing rules, suitability requirements, local registration triggers).
- Under-scoped ongoing obligations (periodic reporting, investor communications, corporate actions).
- Tax frictions that erode the added value thesis (withholding, reporting, complexity costs).

B.4 Phase 3 — Build, integration & vendor onboarding

Case-specific focus for this phase: The current custody design indicates regulated custodian. Its precise legal and operational role remains to be confirmed. The current transfer environment indicates ATS or MTF; this does not establish a local legal venue classification. The current settlement design indicates T+1/T+2 securities settlement. Its operational and legal fit remains to be validated. Transfer-rule implementation must reflect: Whitelist + EU gating + 12-month lock-up.

Objectives

- Finalize the technology stack and integrate custody, settlement, onboarding, and reporting workflows.
- Implement transfer rules and lifecycle event handling (issuance, redemptions, distributions, corporate actions).
- Establish operational readiness (controls, monitoring, incident response, reconciliation).

Key decisions

- Token standard / ledger choice and compatibility with custody and venue requirements.
- Settlement model (atomic delivery-versus-payment vs. coordinated settlement through intermediaries).
- Data and reporting architecture (who stores what; audit trail; privacy boundaries).
- Outsourcing vs. in-house responsibility split (transfer agent, admin, reporting).

Deliverables (checklist)

- Final technical architecture and integration plan (custody, venue, payments, identity/KYC).
- Rule engine / whitelist implementation and test suite (including jurisdictional gates).
- Operational runbooks (reconciliation, incident response, corporate actions, investor support).
- UAT results and sign-offs (issuer, custodian, venue, admin/TA as relevant).
- Security review (key management model, access control, change management, monitoring).

Primary stakeholders

- Issuer operations/IT, tokenization platform/tech provider, regulated custodian where explicitly selected.
- Selected transfer-environment integration team, payment rails (if applicable), admin/TA providers.
- Compliance and risk functions for control design and approval.

Dependencies / gates

- Signed vendor contracts, SLAs, and responsibility matrix (RACI) across the stack.
- Completed integration testing and reconciliation proof (end-to-end lifecycle flows).
- Readiness gate: controls, monitoring, and support operations in place.

Risk hotspots

- Integration complexity and reconciliation gaps across custody/venue/admin systems.
- Rule enforcement drift (transfer restrictions not consistently applied across venues/wallets).
- Operational single points of failure (provider concentration, key management, support load).

B.5 Phase 4 — Issuance, distribution & launch

Case-specific focus for this phase: Any legally required launch permissions and onboarding constraints must be independently confirmed for: Germany, Netherlands. Intended investor group for launch planning: professional investors, other eligible non-retail investors (local classification unconfirmed). Eligibility under local law must be confirmed. The launch transfer environment currently indicates ATS or MTF; this does not establish a local legal venue classification. The current liquidity intent is quarterly transfer windows. No market status, matching demand or liquidity support is inferred.

Objectives

- Execute issuance mechanics and investor onboarding under the independently verified offering pathway.
- Operationalize disclosures, subscriptions, allocations, and post-issuance record-keeping.
- Operationalize only the user-stated and legally validated transfer process, if any; no venue listing, market making or liquidity support is assumed.

Key decisions

- Distribution channel mix (direct, intermediated, platform marketplace, placement agents) as legally validated.
- Subscription mechanics and settlement timing (funding rails, DVP, cutoff times) as actually specified.
- Transfer model, if any: stated approvals, eligibility, timing and limitations; no cadence or support mechanism is assumed.

Deliverables (checklist)

- Final investor communication pack and distribution checklist.
- Subscription and allocation runbook; reconciliation and exception handling.
- Go-live monitoring plan and incident escalation paths.
- Documented transfer-process setup, if applicable, and evidence-based monitoring metrics.

Primary stakeholders

- Issuer distribution team, platform operator, onboarding/KYC provider.
- Custodian, settlement and transfer providers explicitly selected for the stated process.
- Legal/compliance for marketing controls and ongoing disclosure obligations.

Dependencies / gates

- Approved marketing/distribution materials and compliance sign-off.
- End-to-end dry run of subscription, allocation, and post-issuance reporting.
- Operational readiness gate: support, monitoring, and governance in place.

Risk hotspots

- Distribution frictions (slow onboarding, suitability issues, cross-border marketing constraints).
- Unsupported transfer or liquidity expectations relative to the actual process and evidence.
- Operational load spikes (investor support, reconciliation exceptions, reporting deadlines).

B.6 Phase 5 — Post-issuance operations & monitoring

Case-specific focus for this phase: Ongoing governance and evidence trails must remain defensible under Luxembourg. Post-issuance compliance must continue to support holders across: Germany, Netherlands. Ongoing operations must reflect the regulated custodian design once its precise role has been independently confirmed. Post-launch monitoring must reflect the actual validated transfer process currently described as ATS or MTF.

Objectives

- Run ongoing compliance, reporting, and investor communications through the instrument’s lifecycle.
- Operate lifecycle events (distributions, redemptions, corporate actions) with strong controls and auditability.
- Monitor provider performance, costs, and risk posture; adjust as regulations and market conditions evolve.

Key decisions

- Governance cadence (board/committee oversight, change approvals, vendor reviews).
- Reporting scope and frequency (investor reporting, regulatory reporting, audit readiness).
- Contingency planning (provider exits, custody changes, venue strategy shifts).

Deliverables (checklist)

- Ongoing reporting calendar and control framework (incl. audits and reconciliations).
- Lifecycle event runbooks and evidence retention procedures.
- Vendor performance KPIs and incident log; periodic risk reviews.
- Contingency and change-management playbook (custody/venue migrations, rule updates).

Primary stakeholders

- Issuer operations, compliance, finance; custodian; admin/TA (where applicable).
- Transfer and settlement providers explicitly retained in scope, if any.
- External auditors/advisers for periodic reviews and regulatory updates.

Dependencies / gates

- Stable operating model and evidence trail sufficient for audit/regulatory scrutiny.
- Clear change-management process for token rules, disclosures, and provider configuration.
- Budgeting and resourcing for ongoing operations (often underestimated in early planning).

Risk hotspots

- Compliance drift as rules change or distribution expands across jurisdictions.
- Operational fragility under low-liquidity or stress scenarios (settlement failures, disputes).
- Provider concentration risk and cost creep relative to value delivered.

Appendix C — TVAC Provider Directory (illustrative)

This appendix is an illustrative reference layer intended to help the reader think about which types of external partners may be relevant in a tokenization project.

It is not a recommendation list, procurement shortlist, or ranking. Inclusion or omission does not imply endorsement, criticism, quality judgment, regulatory approval, or case suitability. Any provider mentioned here must be assessed independently for legal fit, jurisdictional coverage, operating model fit, technical capability, conflicts, cost, and execution reliability. This appendix is non-exhaustive and subject to change.

C1. How to use this appendix

Use this appendix to identify the provider categories a case may require, not as a signal that a specific provider should be engaged.

In many cases, the main value of the directory is not the names themselves, but the discipline of asking:

- Which functions must be covered?
- Which functions can be combined?
- Which functions must remain regulated or jurisdiction-specific?
- Which dependencies create concentration risk or vendor-capture risk?
- Which parts of the stack are critical before launch, and which can wait?

C2. Typical provider categories

Depending on the case, a tokenization project may require some or all of the following partner categories:

- legal & regulatory advisers
- tax advisers
- issuance / tokenization platform
- smart-contract / token engineering support
- custody / wallet infrastructure
- transfer agent / registry administration
- fund administration or SPV administration
- trading venue / ATS / MTF / exchange partner
- transfer or liquidity-support provider, only where explicitly required and legally permitted
- KYC / AML / onboarding provider
- payment / settlement / banking partner
- audit / assurance / cybersecurity reviewers
- valuation, data, or oracle-related providers
- distribution / placement / investor onboarding partners
- systems integration / workflow automation support

C3.1 Legal & regulatory advisers

These advisers help determine whether the proposed structure is legally enforceable, properly classified, and consistent with the intended issuance, distribution, transfer, custody, and disclosure model.

Typical use cases:

- wrapper design
- securities-law classification
- cross-border marketing analysis
- transfer restriction design
- offering documentation
- legal opinions and regulatory scoping

C3.2 Issuing platforms

These providers typically support issuance workflows, lifecycle management, permissions, investor access rules, and parts of the reporting or compliance stack.

Typical use cases:

- token issuance
- lifecycle events
- permissions / whitelisting
- cap-table or investor-position management
- compliance rule integration
- reporting workflows

C3.3 Custody, trading & settlement

These providers support safekeeping, wallet infrastructure, execution, venue access, settlement, and post-trade operations.

Typical use cases:

- institutional custody
- secure wallet infrastructure
- settlement support
- exchange / venue access
- post-trade servicing
- collateral mobility or transfer control support

C4. Practical selection principles

When evaluating providers, the reader should typically assess:

- jurisdictional fit
- regulatory status where relevant
- experience with the relevant asset class
- operating-model compatibility
- integration burden
- exit options / portability
- fee model
- service resilience
- auditability and evidence quality
- concentration risk

Where possible, provider choices should reinforce the case logic rather than undermine it. A weak provider stack can easily erode the value thesis of an otherwise promising structure.

C5. Important caution

The right provider stack is highly case-dependent.

A provider suitable for one asset class or operating model may not fit another. A provider strong in digital-asset custody may not be the right choice for transfer administration or legal structuring. A venue that looks attractive commercially may create regulatory or operating-model problems if the case does not actually need that level of transferability.

Appendix D — Glossary & abbreviations

A practical reference for the most common terms and abbreviations used in this report. Spelling note: tokenization uses “z”.

Term	Meaning
Added Value (AV)	TVAC’s net assessment of whether tokenization appears to improve a case overall once both upside and friction are weighed together. Formula: AV = (NO + CS + RR) – (TC + NR).
AML	Anti-Money Laundering controls intended to detect, prevent, and report suspicious financial activity.
Asset owner / Issuer	The originating party, sponsor, manager, or issuing structure responsible for launching or sponsoring the tokenized case being assessed.
Cash leg	The cash side of a transaction or settlement flow, as distinct from the asset or token side.
Conditional Go	A Tokenization Value Verdict indicating that tokenization may add net value, but remains materially dependent on design discipline, implementation choices, and resolution of key gating conditions.
Custody	The safeguarding, control, and recordkeeping of assets or positions on behalf of investors or market participants.
Distribution scope	The intended investor reach of an offering or tokenized structure, including geography, investor type, and access conditions.
DvP	Delivery versus Payment. A settlement model in which transfer of the asset and transfer of payment are linked so that one does not occur without the other.
Economic rights	The financial entitlements attached to a token or instrument, such as income participation, coupon, principal repayment, or profit-sharing.
Eligibility controls	Rules and mechanisms that determine who may acquire, hold, or transfer a token or instrument.
ESG	Environmental, Social, and Governance factors.
Execution risk	The risk that a concept that appears viable in theory fails in practice because implementation, coordination, economics, or governance proves weaker than assumed.
Governance	The decision-making, control, accountability, and escalation framework surrounding a structure, platform, issuer, or operating model.
Investor segmentation	The way the target investor base is defined and limited by investor type, sophistication, jurisdiction, or eligibility criteria.
KYC	Know Your Customer procedures used to verify customer identity and onboarding eligibility.
Liquidity window	A predefined period or mechanism allowing limited secondary trading, transfer, redemption, or exit opportunities.
MiCA	Markets in Crypto-Assets Regulation in the European Union.
MiFID II	Markets in Financial Instruments Directive II in the European Union.
MTF	Multilateral Trading Facility under EU regulatory frameworks.
Net Added Value	See Added Value (AV).
New Opportunities (NO)	Case-specific upside that tokenization creates or materially strengthens, such as new investor reach, new operational utility, collateral use, programmability, or improved strategic flexibility.
New Risks (NR)	Additional risks introduced by tokenization, including legal, regulatory, operational, technological, governance, distribution, or market-structure risks.
NO / CS / RR / TC / NR	Shorthand for TVAC’s five factors: New Opportunities, Cost Savings, Risk Reduction, Tokenization Costs, and New Risks.
Non-viable	A Tokenization Value Verdict indicating that the current structure does not support a positive tokenization value case under the TVAC methodology and should not proceed without substantial redesign.
Professional investor	An investor category used in many jurisdictions for parties assumed to have greater sophistication, capacity, or experience than retail investors.
Prospectus	A formal disclosure document that may be required for certain public offerings or investor solicitations under applicable law.

Term	Meaning
Provider economics / value-capture misalignment	The risk that provider fees, integration burden, contractual dependencies or operating responsibilities absorb or displace the measurable value that the initiating institution is expected to capture from the tokenized workflow.
Registry	The official or operational record of holdings, ownership, or investor positions.
Retail investor	A non-professional investor who generally receives higher levels of regulatory protection than professional or institutional investors.
Rights mapping	The process of making clear what legal, economic, governance, and transfer rights a token actually represents and how those rights are enforceable.
Risk Reduction (RR)	The extent to which tokenization reduces existing risk in the conventional setup, for example through better controls, improved transparency, clearer auditability, or lower reconciliation burden.
Self-custody	A model in which the end holder directly controls the relevant wallet, keys, or token position without relying on a traditional third-party custodian.
Settlement model	The way asset transfer, cash transfer, record updates, and finality are operationally achieved.
SPV	Special Purpose Vehicle. A ring-fenced legal entity often used for issuance, structuring, asset isolation, or liability segregation.
Tokenization Costs (TC)	The additional costs introduced by tokenization, including structuring, legal work, compliance, integration, platform costs, governance overhead, servicing changes, and operating complexity.
Transfer controls	Rules, permissions, and technical or legal mechanisms governing who may transfer, receive, or hold the token.
Transfer restrictions	Specific limitations placed on secondary transfers, often for regulatory, contractual, or investor-eligibility reasons.
TVAC	Tokenization Value-Add Calculator. A structured decision-support methodology for assessing whether tokenization appears to create net added value in a specific case.
Vendor-capture risk	The risk that a structure becomes economically or operationally overly dependent on one platform, provider, or implementation path in a way that shifts value away from the initiating party.
Viable	A Tokenization Value Verdict indicating that the TVAC methodology identifies a positive tokenization value case under the described structure, subject to validation and disciplined execution. It is not an investment, credit, yield, valuation, legal, tax, or suitability assessment.
Whitelist / Whitelisting	A mechanism for restricting participation or transferability to pre-approved investors, wallets, counterparties, or participants.

Appendix E — Case inputs used for assessment

E.1 Full verbatim case description

Full original free-text case input provided by the user.

We are a privately owned real estate holding company owning three stabilized residential assets in Berlin and Hamburg through German GmbHs. The properties generate steady rental income but require energy upgrades, façade improvements, and selected tenant-area refurbishments to remain competitive and compliant with evolving ESG standards.

We intend to raise EUR 15–20 million to finance renovations, partially refinance bank debt, and strengthen liquidity buffers. Instead of raising additional traditional mezzanine capital, we are evaluating tokenization through a Luxembourg SPV issuing profit-participating notes linked to net rental cash flow after operating costs and senior debt service.

The tokens would represent clear contractual economic rights, including profit participation and principal repayment, but no operational control over the properties. We would target professional and accredited investors in Germany and the Netherlands only, using a whitelist-based onboarding process with full KYC/AML.

We plan a 12-month lock-up followed by structured quarterly liquidity windows on a regulated venue. Custody would be provided by a regulated custodian. Settlement follows T+1/2 standards aligned with existing capital market workflows. Key considerations include cross-border tax treatment, renovation execution risk, valuation stability, and interest-rate sensitivity. The primary rationale for tokenization is improved investor transparency, fractional access for smaller professional tickets, and long-term administrative efficiency.

E.2 Structured input record

This record reproduces the structured inputs supplied for this assessment. For dropdowns and checkboxes, it also shows the alternatives that were available but not selected.

Field	User response	Other available options not selected
Issuance jurisdiction	Luxembourg	Not applicable
Sale jurisdictions (comma-separated)	Germany, Netherlands	Not applicable
Instrument type	Digital wrapper (CSD/TA/registrar)	Sole-record / native tokenized security
Custody model	Regulated custodian	Transfer agent / CSD; Self-custody (wallets)
Secondary-market liquidity	Quarterly windows	None; Ad-hoc OTC; Continuous
Secondary-market venue	ATS / MTF	DEX (permissioned); OTC; CSD / TA-registered
Payment / cash-leg model	T+1/T+2 securities settlement	Not decided yet; Not material at this stage; Traditional bank / fiat payments; Custodian / bank / TA / CSD settlement; Stablecoin / tokenized cash; Deposit token / tokenized deposit / institutional DvP rail; Advanced: T+0 / atomic DvP; Advanced: off-chain booking
Offering disclosure	Exempt / limited-offering route (jurisdiction-dependent)	KID (PRIIPs); Prospectus
Transfer restrictions	Whitelist + EU gating + 12-month lock-up	Not applicable
Tax considerations (optional)	Cross-border withholding considerations	Not applicable
Investor types	Professional; Qualified / accredited (jurisdiction-dependent)	Retail

An option shown as not selected was not necessarily actively considered or rejected by the user; it was simply not selected for this assessment.

> **Scenario comparison:** Alternative options shown here can help identify design variables worth testing in a new assessment or in TVAC Scenario Lab—for example a different custody model, settlement model, participant type or transfer setup.

Appendix F — Disclaimer

Disclaimer

This report is provided for informational and decision-support purposes only. It does not constitute legal, regulatory, tax, accounting, investment, valuation, or financial advice, and it should not be relied upon as a substitute for professional advice tailored to the specific facts and circumstances of the case.

TVAC is a structured assessment tool designed to evaluate whether and when tokenization may create added value relative to conventional alternatives. Its outputs reflect a methodology-based analytical framework operating under uncertainty. They are not guarantees of commercial success, feasibility, investor demand, regulatory approval, liquidity, execution quality, budget outcome, or risk reduction.

Any scores, verdicts, scenarios, stakeholder assessments, suggested actions, implementation observations, and roadmap elements in this report are indicative only and depend on the quality, completeness, and interpretation of the input data, as well as on assumptions about structure, jurisdiction, service-provider readiness, market conditions, execution capability, and timing.

Tokenization outcomes can differ materially from this assessment if key conditions change, including legal structuring, investor scope, jurisdiction count, custody model, distribution pathway, settlement design, governance arrangements, servicing readiness, technology integration, cost allocation, evidence quality, or liquidity support.

A TVAC verdict is a tokenization added-value verdict only. It must not be presented or used as investment approval, credit rating, yield validation, issuer endorsement, regulatory approval, due diligence report, or investor recommendation. TVAC does not validate projected returns, coupon payments, credit quality, valuations, legal enforceability, tax treatment, or investor suitability.

This report does not replace formal legal analysis, regulatory review, tax analysis, technical architecture review, cybersecurity review, accounting assessment, operational diligence, commercial validation, or board-level decision processes. Independent advice should always be obtained before any structuring, issuance, distribution, investment, operational, or implementation decision is made.

Neither TVAC nor Astoval assumes responsibility for actions taken on the basis of this report alone. Users remain solely responsible for all commercial, legal, regulatory, operational, and strategic decisions related to any proposed tokenization initiative.

TVAC

Tokenization Value-Add Calculator

A PRODUCT BY ASTOVAL

Astoval provides independent, AI-enhanced early-stage decision support for asset tokenization, helping organizations assess whether tokenization creates genuine added value before significant resources are committed to legal structuring, platform selection or implementation.

PRIVACY AND DATA

Case inputs and generated reports are processed for delivery and are not intentionally retained as a persistent customer-case record in the standard self-service flow. OpenAI API data is not used for model training unless the API customer opts in; separate abuse-monitoring retention may apply for up to 30 days.

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